

Phase 4 Formal Framework: Decoupled Planning and Learning Burdens in Symbolic Action Models

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Abstract

This document establishes the formal theoretical framework, research objectives, research questions, hypotheses, algorithms, and statistical analysis protocol for Phase 4 of the research program: *Mechanic-Induced Planning-Model Burden Decomposition*. We formalize the distinction between Planning Burden (H_P) and Learning Burden (B), formulate the Decoupling Hypothesis and Zone B (the critical blindspot where models appear accurate under passive evaluation yet induce catastrophic planning failure), present complete algorithmic pseudocode, specify an 8-domain International Planning Competition (IPC) empirical matrix, and detail the statistical testing pipeline.

1 Context, Problem Statement, and Research Gaps

1.1 Discrete Symbolic Planning vs. Continuous Control World Models

In continuous deep world models (e.g., DreamerV3, PlaNet), state representations lie in metric spaces ($\mathbb{R}^d$), and dynamics prediction errors typically manifest as gradual drift measured by mean squared error (MSE) across rollout horizons. In contrast, **symbolic planning environments possess discrete, combinatorial topology governed by first-order logic constraints**.

A single omitted precondition in an action schema does not cause a continuous numerical perturbation; rather, it performs a discrete topological relaxation on the state-space reachability graph $\mathcal{R}(s_0)$. This creates *phantom transition shortcuts* that a forward search planner greedily exploits, causing catastrophic real-world execution failure.

1.2 The Three-Tier Problem Statement

1. **Tier 1 (Known Phenomenon):** High passive transition prediction accuracy ($A_{\text{pred}} \geq 98\%$) does not guarantee planning success (*Verified-vs-Correct Gap*, Aguilar Martín et al., 2026).
2. **Tier 2 (The Unformalized Gap):** Prior literature (PDDLgym, 2020; TCP, 2026) observed runtime and sample complexity variation across planning domains, but **there is no formal framework to decompose and quantify** planning search difficulty (H_P) independently from model-learning sample complexity (B).
3. **Tier 3 (The Decoupling Frontier):** No formal theory or empirical benchmark has demonstrated that H_P and B can be **causally decoupled** under minimal rule interventions—meaning that specific operator modifications can leave planning search tractable while rendering passive model learning infinitely inadequate.

2 Research Objectives (RO) and Research Questions (RQ)

2.1 Research Objectives (RO)

- **RO1 (Formalization):** Formalize Planning Burden H_P and Learning Burden B as independent, mathematically well-defined computational estimands.
- **RO2 (Empirical Decoupling & Zone B Existence):** Empirically prove the existence of **Zone B** (tractable planning burden H_P , but divergent/infinite learning burden B) across 8 diverse IPC planning domains under single-precondition interventions.
- **RO3 (Diagnostic Benchmark Synthesis):** Formulate an automated diagnostic benchmark protocol that synthesizes minimal rule interventions to detect Zone B vulnerabilities before deployment.

2.2 Research Questions (RQ)

- **RQ1 (Burden Decoupling):** Do marginal changes in planning burden (ΔH_P) and learning burden (ΔB) causally decouple ($|\rho| < 0.30$) when isolated operator preconditions are modified?
- **RQ2 (Zone B Blindness):** Is Zone B the primary mathematical driver of passive evaluation failure in action model learning?
- **RQ3 (Diagnostic Benchmark Generation):** Can minimal rule interventions systematically synthesize test instances that expose Zone B failures where passive accuracy remains $A_{\text{pred}} = 1.000$?

3 Formal Definitions and Theoretical Foundations

Let $M^* = \langle \mathcal{P}, \mathcal{A}, \mathcal{T}^* \rangle$ denote the ground-truth STRIPS domain, where $\mathcal{P}$ is the set of first-order predicates, $\mathcal{A}$ is the set of action schemas $a = \langle \text{Pre}(a), \text{Add}(a), \text{Del}(a) \rangle$, and $\mathcal{T}^*$ represents true state transitions. Let $\mathcal{X} = \langle s_0, S_g \rangle$ be a planning problem instance.

Definition 1 (Planning Burden H_P). Given an executable domain model M , problem instance $x = \langle s_0, S_g \rangle$, and planner P , the **Planning Burden** is defined as:

$$H_P(M, x) \triangleq \log_2(1 + N_{\text{exp}}(P, M, x)) \quad (1)$$

where $N_{\text{exp}}(P, M, x)$ is the exact number of state nodes expanded by planner P to return a valid plan π_M (or determine unsolvability).

Definition 2 (Play Regret R_{play}). Given a candidate model $\widehat{M}$, ground-truth domain M^* , and instance x , let $\pi_{\widehat{M}}$ be the plan synthesized by planner P on $\widehat{M}$. The **Play Regret** of executing $\pi_{\widehat{M}}$ in true physics M^* is:

$$R_{\text{play}}(\pi_{\widehat{M}}, M^*, x) \triangleq \begin{cases} \text{cost}_{M^*}(\pi_{\widehat{M}}) - \text{cost}_{M^*}(\pi_{M^*}^*) & \text{if } \pi_{\widehat{M}} \text{ executes successfully to reach } S_g \text{ in } M^*, \\ \infty & \text{if execution violates any } \text{Pre}^*(a_t) \text{ or fails to reach } S_g. \end{cases} \quad (2)$$

Definition 3 (Learning Burden B). Given an inductive learner L , training trace distribution D generated by valid executions on M^* , planner P , and regret tolerance $\varepsilon \geq 0$, the **Learning Burden** is:

$$B_{L,D,P,\varepsilon}(M^*, x) \triangleq \min \{n \in \mathbb{N} : \mathbb{E}_{T \sim D^n} [R_{\text{play}}(P(L(T)), M^*, x)] \leq \varepsilon\} \quad (3)$$

If no finite sample size n guarantees expected regret $\leq \varepsilon$, then $B = \infty$.

Definition 4 (Burden Decoupling). Let $\Delta_m H_P = H_P(M^* \setminus \{m\}, x) - H_P(M^*, x)$ and $\Delta_m B = B(M^* \setminus \{m\}, x) - B(M^*, x)$ be the marginal shifts induced by mechanic intervention m . The burdens are **decoupled** if their rank correlation across interventions satisfies:

$$|\rho_{\text{Spearman}}(\Delta H_P, \Delta B)| < 0.30 \quad (4)$$

Definition 5 (The Four Hardness Zones). The $(\Delta H_P, \Delta B)$ plane partitions into four distinct operational quadrants:

- **Zone A (Tractable / Robust):** Low H_P , Low B .
- **Zone B (Dangerous Blindspot):** Low/Normal H_P , Divergent/Infinite B ($R_{\text{play}} = \infty$).
- **Zone C (Planning-Heavy / Model-Light):** High H_P , Low B .
- **Zone D (Intrinsically Hard):** High H_P , High/Infinite B .

4 Propositions and Proofs

Proposition 1 (Phantom Edges and Passive Blindness). Let M^* be the ground truth domain. Suppose $\widehat{M}$ omits a necessary precondition $p \in \text{Pre}^*(a)$ such that $\text{Pre}(\widehat{a}) = \text{Pre}^*(a) \setminus \{p\}$ while $\text{Eff}(\widehat{a}) = \text{Eff}^*(a)$.

1. On any observational trace distribution $D \sim M^*$, every transition (s, a, s') satisfies $s \models \text{Pre}^*(a) \implies s \models \text{Pre}(\widehat{a})$. Consequently, passive transition prediction accuracy is identically:

$$A_{\text{pred}}(\widehat{M}) = 1.000 \quad (5)$$

2. The state reachability graph satisfies $\mathcal{R}(s_0; \widehat{M}) \supseteq \mathcal{R}(s_0; M^*)$. If the optimal plan in $\widehat{M}$ traverses an edge enabled by $s \not\models p$, then $\pi_{\widehat{M}}$ fails on step 1 of that edge in M^* , yielding:

$$R_{\text{play}}(\pi_{\widehat{M}}, M^*, x) = \infty \implies B_{L,D,P,\varepsilon} = \infty \quad (6)$$

Proof. In $D \sim M^*$, all recorded transitions originate from valid executions on M^* . Hence, $\forall (s_t, a_t, s_{t+1}) \in D$, $s_t \models \text{Pre}^*(a_t)$ holds. Since $\text{Pre}(\widehat{a}_t) \subset \text{Pre}^*(a_t)$, it follows trivially that $s_t \models \text{Pre}(\widehat{a}_t)$. Because $\text{Eff}(\widehat{a}_t) = \text{Eff}^*(a_t)$, the predicted state $\widehat{s}_{t+1} = (s_t \setminus \text{Del}(\widehat{a}_t)) \cup \text{Add}(\widehat{a}_t) = s_{t+1}$ matches the true state with zero error, establishing $A_{\text{pred}} = 1.000$.

However, in planning on $\widehat{M}$, forward search explores the relaxed state graph $G(\widehat{M})$. The relaxation adds directed edges $E_{\text{phantom}} = \{(s, s') : s \models \text{Pre}(\widehat{a}) \wedge s \not\models p\}$. If any such edge provides a shortcut towards S_g , the optimal planner selects it. Executing this step on physical domain M^* tests $s \models \text{Pre}^*(a)$, which fails because $s \not\models p$. By Definition 2, execution halts with $R_{\text{play}} = \infty$. Because positive traces from D never supply counterfactual evidence of illegal actions, an inductive learner cannot recover p , forcing $B = \infty$. $\square$

5 Algorithmic Pseudocode

6 Expanded Multi-Domain Experiment Matrix (8 IPC Domains)

The empirical matrix expands across **8 representative IPC domains**, ensuring topological and structural diversity:

Algorithm 1 Computation of Planning Burden H_P

Require: Domain model M , Problem $x = \langle s_0, S_g \rangle$, Forward Planner P , Node cutoff $N_{\max}$

Ensure: Planning burden H_P , Plan π_M , Node count $N_{\exp}$

```
1: Initialize search queue  $Q \leftarrow [s_0]$ , visited set  $V \leftarrow \{s_0\}$ ,  $N_{\exp} \leftarrow 0$ 
2: while  $Q$  is not empty and  $N_{\exp} < N_{\max}$  do
3:   Pop state  $s \leftarrow Q.\text{pop}()$ ,  $N_{\exp} \leftarrow N_{\exp} + 1$ 
4:   if  $S_g \subseteq s$  then
5:     return  $\log_2(1 + N_{\exp})$ , reconstruct plan  $\pi_M$ ,  $N_{\exp}$ 
6:   end if
7:   for ground action  $a$  applicable in  $s$  under  $M$  do
8:      $s' \leftarrow \text{apply}(s, a, M)$ 
9:     if  $s' \notin V$  then
10:       $V.\text{add}(s')$ ,  $Q.\text{push}(s')$ 
11:    end if
12:  end for
13: end while
14: return  $\log_2(1 + N_{\exp})$ , None,  $N_{\exp}$ 
```

Algorithm 2 Computation of Learning Burden B

Require: Learner L , Ground truth M^* , Problem x , Trace generator D , Planner P , Regret threshold ε , Sample sizes $\{n_1, \dots, n_k\}$, Seeds K

Ensure: Learning burden $B \in \mathbb{N} \cup \{\infty\}$, Regret curve

```
1: for  $n \in \{n_1, \dots, n_k\}$  do
2:   regrets  $\leftarrow []$ 
3:   for seed  $\in \{1, \dots, K\}$  do
4:     Traces  $T \leftarrow D.\text{sample}(n, \text{seed})$ 
5:     Learned model  $\widehat{M} \leftarrow L.\text{learn}(T)$ 
6:     Plan  $\pi \leftarrow P.\text{solve}(\widehat{M}, x)$ 
7:     Success, cost  $\leftarrow \text{execute\_plan}(\pi, M^*, x)$ 
8:     if Success then
9:       regrets.append(cost - optimal_cost( $M^*$ ,  $x$ ))
10:    else
11:      regrets.append( $\infty$ )
12:    end if
13:  end for
14:  if mean(regrets)  $\leq \varepsilon$  then
15:    return  $n$ , recorded regret curve
16:  end if
17: end for
18: return  $\infty$ , recorded regret curve
```

7 Statistical Analysis Protocol

1. **Decoupling Hypothesis Test:** Spearman's rank correlation $\rho(\Delta H_P, \Delta B)$ is calculated over all intervention conditions. The null hypothesis $H_0 : |\rho| \geq 0.30$ is tested against $H_1 : |\rho| < 0.30$ at significance level $\alpha = 0.05$.
2. **Bootstrap Confidence Intervals:** 95% confidence intervals for ρ , ΔH_P , and ΔB are derived using 1,000 bootstrap resamples across problem seeds.
3. **Effect Size Analysis:** Cohen's d is computed for shifts in expanded nodes ($\Delta N_{\exp}$) and

Domain	Topological Class	Bottleneck Mechanism	Precondition Interventions
Blocksworld	Relational Stacking	Spatial clearance, hand capacity	Omit <code>clear</code> , <code>ontable</code> , <code>holding</code>
Gripper	Resource Bottleneck	Two-handed carrying capacity	Omit <code>free</code> , <code>at-robby</code> , <code>carry</code>
Ferry	Vehicle Routing	Single-car capacity constraint	Omit <code>empty-ferry</code> , <code>at-ferry</code> , <code>o</code>
Miconic	Passenger Elevators	Floor ordering, boarding constraints	Omit <code>lift-at</code> , <code>boarded</code> , <code>above</code>
Driverlog	Multi-Vehicle Logistics	Truck boarding, path connectivity	Omit <code>empty</code> , <code>at</code> , <code>driving</code>
Hanoi	Recursive Depth	Strict disk ordering ($o_1 < o_2$)	Omit <code>smaller</code> , <code>clear</code>
Satellite	Sensor Scheduling	Sensor calibration, instrument pointing	Omit <code>calibrated</code> , <code>pointing</code> , <code>po</code>
Rovers	Energy & Store-Forward	Battery, sample store, line-of-sight	Omit <code>empty</code> , <code>at</code> , <code>available</code>

Table 1: The 8 IPC Benchmark Domains and Target Interventions for Phase 4.

sample complexity (ΔB) relative to baseline M^* .

4. **Quadrant Scatter Analysis:** Every intervention is projected onto the $(\Delta H_P, \log_{10} B)$ coordinate plane with median dividers to empirically delineate Zones A, B, C, and D.

8 Core Contributions and Scoped Limitations

8.1 Packaged Contributions

- **C1 (Burden Decomposition Framework):** The first formal framework defining Planning Burden (H_P) and Learning Burden (B) as orthogonal, mathematically decoupled estimands.
- **C2 (Theoretical Proof of Decoupling & Zone B):** Theoretical proof (Proposition 1) and empirical demonstration that passive predictive accuracy (A_{pred}) fails to detect Zone B planning collapse.
- **C3 (8-Domain Diagnostic Benchmark):** An open-source, reproducible diagnostic benchmark spanning 8 IPC domains that systematically isolates phantom shortcuts across diverse symbolic learners (FAMA, LOCM2, FastLAS).

8.2 Explicit Limitations

- **Discrete STRIPS Scope:** The current formalization is restricted to discrete, deterministic STRIPS domains; continuous state variables and stochastic transitions require future hybrid extensions.
- **Planner Sensitivity:** H_P is defined relative to state-space forward search (BFS/A*); while relative ordering remains robust, exact node counts vary under alternative planning paradigms (e.g., SAT/symbolic planners).
- **Passive Trace Assumption:** Learning burden B is analyzed under positive observational traces; active counterfactual query synthesis is left to Paper 2.